

Instructions Shape **Production** of Language, not **Processing**

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[Paper](#)

[Code](#)

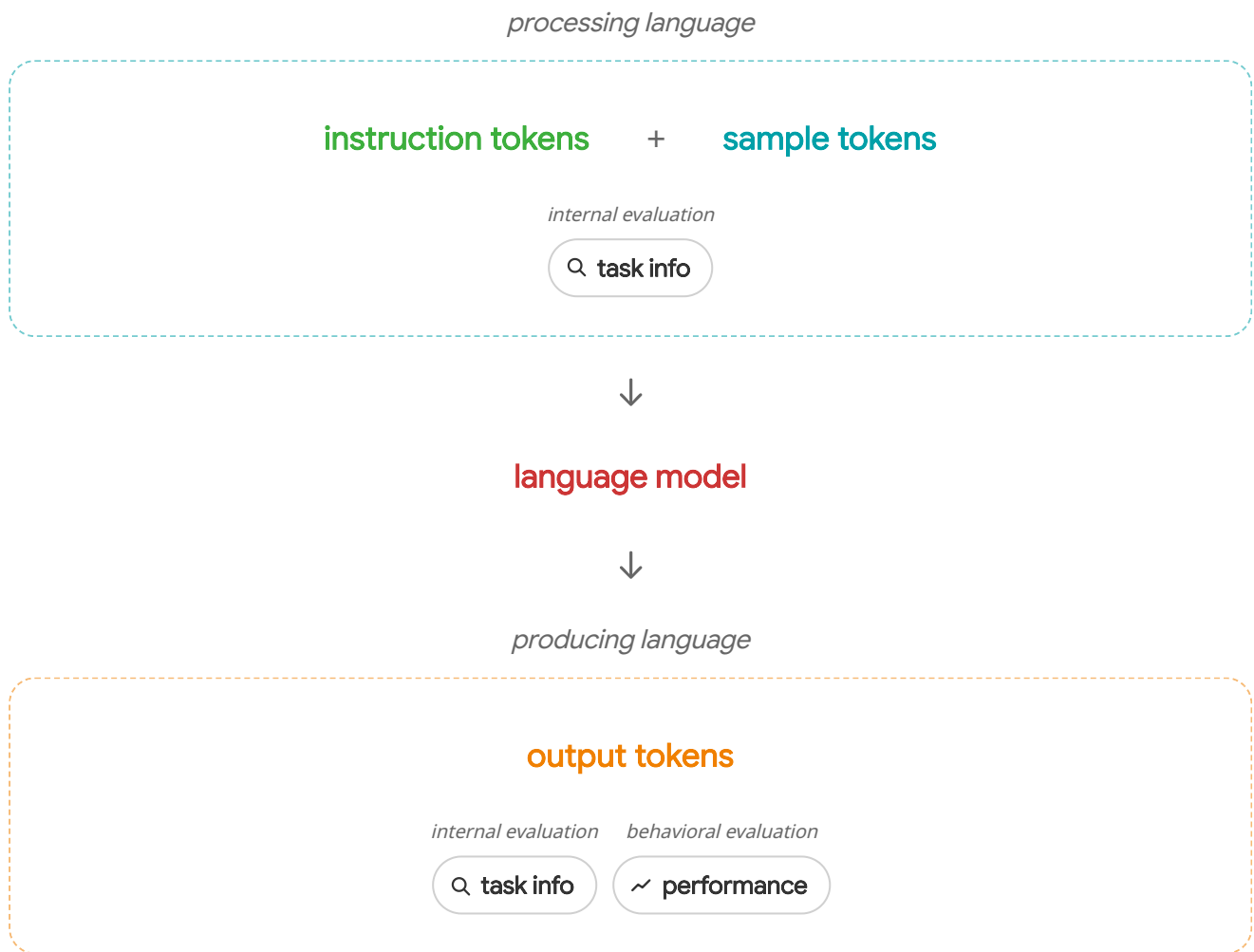
Overview

Instructions trigger a **production-centered mechanism**.

Task-specific information in **sample tokens** is stable across prompt variations and aligns only weakly with task performance, while the same information in **output tokens** varies substantially and is correlated with performance.

Inspired by cognitive theories of language, we distinguish two stages: **processing**, where the model encodes input, and **production**, where it generates a response. We use token position as a proxy: representations at **sample tokens** reflect processing; representations at **output tokens** reflect production. We do **not** probe instruction tokens. At each layer, we probe α how much task-specific information the model carries at sample and output positions, how that information changes

across prompting variations, and how it responds to attention-based interventions.



See the tasks and models we use →

More on the experimental setup →

Project page · May 2026